(*) Advanced LOOKUPS - INDEX And MATCH Functions:-

INDEX and MATCH are advanced Excel Lookup functions used together to find and retrieve a specific value from a table.

- INDEX: Returns a value from a particular row & column in a selected range.
- MATCH: Finds the position of a specific value within a row or column.

(*) When used together - Match finds the position of the required value, and INDEX returns the corresponding result. So together they provide a flexible way to perform advanced lookups in Excel.

(*) INDEX Syntax - =INDEX (array, row_num, [column_num]).

array - The range containing the data.
row_num - The row position of the value.
column_num - The column position (optional for a single-column range).

(*) MATCH Syntax - =MATCH (lookup_value, lookup_array, [match_type]).

lookup_value - The value you want to find.
lookup_array - The range where you want to search.
match_type - Use 0 for an exact match.

(*) Example using INDEX and MATCH together:-

Employee_ID	Employee Name	Salary
101	Priya	25,000
102	Rahul	30,000
103	Anu	28,000
104	Kiran	35,000

Q:- Find the Salary of Rahul using his Employee name.

=INDEX(C2:C5, MATCH("Rahul", B2:B5, 0))

(*) How it works:

- MATCH("Rahul", B2:B5, 0) - finds Rahul's position in the Employee Name column. The result is 2.
- INDEX(C2:C5, 2) - returns the 2nd value from the Salary Column.

Output - 30,000.

(*) Advantages of INDEX and MATCH :-

- Can look up values to the left/right of the lookup column.
- More flexible than traditional VLOOKUP.
- Can work with both vertical & horizontal data.
- Doesn't depend on a fixed column number, so it is more adaptable when columns are inserted or rearranged.

(*) Syntax of INDEX & MATCH Together:

=INDEX(array, MATCH(row_value, row_lookup, 0), MATCH(column_value, column_lookup, 0)).

(*) Example - Suppose your employee data contains this columns

Emp code	Employee Name	City.
1	JP Kumar	Mumbai
4	R Vasu	Chennai
9	Arjun Jain	Mumbai
13	Anupam Mishra	Mysore
17	Satish Puri	Mumbai

Q:- Retreive the employee's city using Emp code & the column heading City.

~~= INDEX (B2 : B :01,~~

~~= INDEX (B2 : $15, MATCH(~~

= INDEX (A2:C6, MATCH(4, A2:A6, 0), MATCH ("City", A1:C1, 0))

o/p - chennai.

- INDEX (A2:C6, ...) - Selects the entire employee data range.
- MATCH (4, A2:A6, 0) - finds the row position of Emp code 4.
- MATCH ("City", A1:C1, 0) - finds the column position of the city heading.
- INDEX - Returns the value of at the intersection - chennai.